



3.3.3 Practical computation of the Householder vector



03 3 3 Practical computation of Householder transformations

fit width

3.3.3.1 The real case

Next, we discuss a slight variant on the above discussion that is used in practice. To do so, we view x as a vector that consists of its first element, x_1 , and the rest of the vector, x_2 . More precisely, partition

$$x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix},$$

where x_1 equals the first element of x and x_2 is the rest of x . Then we will wish to find a Householder vector $u = \begin{pmatrix} 1 \\ u_2 \end{pmatrix}$ so that

$$\left(I - \frac{1}{\tau} \begin{pmatrix} 1 \\ u_2 \end{pmatrix} \begin{pmatrix} 1 \\ u_2 \end{pmatrix}^T \right) \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} \pm \|x\|_2 \\ 0 \end{pmatrix}.$$

Notice that y in the previous discussion equals the vector $\begin{pmatrix} \pm \|x\|_2 \\ 0 \end{pmatrix}$, so the direction of u is given by

$$v = \begin{pmatrix} x_1 \mp \|x\|_2 \\ x_2 \end{pmatrix}.$$

We now wish to normalize this vector so its first entry equals "1":

$$u = \frac{v}{\nu_1} = \frac{1}{x_1 \mp \|x\|_2} \begin{pmatrix} x_1 \mp \|x\|_2 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ x_2/\nu_1 \end{pmatrix}.$$

where $\nu_1 = x_1 \mp \|x\|_2$ equals the first element of v . (Note that if $\nu_1 = 0$ then u_2 can be set to 0.)

3.3.3.2 The complex case (optional)

Let us work out the complex case, dealing explicitly with x as a vector that consists of its first element, x_1 , and the rest of the vector, x_2 . More precisely, partition

$$x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix},$$

where x_1 equals the first element of x and x_2 is the rest of x . Then we will wish to find a Householder vector $u = \begin{pmatrix} 1 \\ u_2 \end{pmatrix}$ so that

$$\left(I - \frac{1}{\tau} \begin{pmatrix} 1 \\ u_2 \end{pmatrix} \begin{pmatrix} 1 \\ u_2 \end{pmatrix}^H \right) \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} \boxed{\pm} \|x\|_2 \\ 0 \end{pmatrix}.$$

Here $\boxed{\pm}$ denotes a complex scalar on the complex unit circle. By the same argument as before

$$v = \begin{pmatrix} x_1 - \boxed{\pm} \|x\|_2 \\ x_2 \end{pmatrix}.$$

We now wish to normalize this vector so its first entry equals "1":

$$u = \frac{v}{\nu_1} = \frac{1}{x_1 - \boxed{\pm} \|x\|_2} \begin{pmatrix} x_1 - \boxed{\pm} \|x\|_2 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ x_2/\nu_1 \end{pmatrix}.$$

where $\nu_1 = x_1 - \boxed{\pm} \|x\|_2$. (If $\nu_1 = 0$ then we set u_2 to 0.)

As was the case for the real-valued case, the choice $\boxed{\pm}$ is important. We choose $\boxed{\pm} = -\text{sign}(x_1) = -\frac{x_1}{|x_1|}$.

3.3.3.3 A routine for computing the Householder vector



The vector

$$\begin{pmatrix} 1 \\ u_2 \end{pmatrix}$$

is the Householder vector that reflects x into $\pm \|x\|_2 e_0$. The notation

$$\begin{bmatrix} \rho \\ u_2 \end{bmatrix}, \tau := \text{Housev} \left(\begin{pmatrix} \chi_1 \\ x_2 \end{pmatrix} \right)$$

represents the computation of the above mentioned vector u_2 , and scalars ρ and τ , from vector x . We will use the notation $H(x)$ for the transformation $I - \frac{1}{\tau} u u^H$ where u and τ are computed by $\text{Housev}(x)$.

Algorithm : $\begin{bmatrix} \rho \\ u_2 \end{bmatrix}, \tau = \text{Housev} \left(\begin{pmatrix} \chi_1 \\ x_2 \end{pmatrix} \right)$	
$\rho = -\text{sign}(\chi_1) \ x\ _2$	$\chi_2 := \ x_2\ _2$
$v_1 = \chi_1 + \text{sign}(\chi_1) \ x\ _2$	$\alpha := \left\ \begin{pmatrix} \chi_1 \\ \chi_2 \end{pmatrix} \right\ _2 (= \ x\ _2)$
$u_2 = x_2 / v_1$	$\rho := -\text{sign}(\chi_1) \alpha$
	$v_1 := \chi_1 - \rho$
	$u_2 := x_2 / v_1$
$\tau = (1 + u_2^H u_2) / 2$	$\chi_2 = \chi_2 / v_1 (= \ u_2\ _2)$
	$\tau = (1 + \chi_2^2) / 2$

Figure 3.3.3.1. Computing the Householder transformation. Left: simple formulation. Right: efficient computation. Note: I have not completely double-checked these formulas for the complex case. They work for the real case.

Remark 3.3.3.2. The function

```
function [ rho, ...
         u2, tau ] = Housev( chi1, ...
                           x2 )
```

implements the function `Housev`. It can be found in

[Assignments/Week03/matlab/Housev.m](#)

Homework 3.3.3.1. Function [Assignments/Week03/matlab/Housev.m](#)

implements the steps in [Figure 3.3.3.1](#) (left). Update this implementation with the equivalent steps in [Figure 3.3.3.1](#) (right), which is closer to how it is implemented in practice.

▼ [Solution](#)

[Assignments/Week03/answers/Housev-alt.m](#)